A CPI-Weighted Web-Based Price Index for Monitoring Inflation and Cost of Living in Türkiye

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***Abstract*—This study proposes a transparent and reproducible framework for monitoring online price dynamics and cost-of-living changes in Türkiye through large-scale web crawling. Daily prices were collected from multiple online sources covering supermarkets, clothing, home goods, construction supplies, cosmetics, technology, rental housing, and a smaller auxiliary set of health-related products. Using these data, a CPI-weighted web-based price index was constructed for the covered categories and compared with reference indicators reported by the Turkish Statistical Institute (TUIK), the Inflation Research Group (ENAG), and the Istanbul Chamber of Commerce (ITO). In addition, nutrition-based hunger-threshold estimates were calculated for a four-person household and compared with the values reported by the Confederation of Turkish Trade Unions (TURK-IS). The results show that web-based estimates provide useful complementary information, although they are sensitive to category coverage, data completeness, online-price availability, and weighting assumptions. Overall, the study demonstrates the potential of transparent web-based price monitoring as a complementary tool for examining inflation dynamics and cost-of-living conditions in Türkiye.**

***Index Terms*—Consumer price index (CPI), cost of living, hunger threshold, inflation measurement, online price index, web crawling, web-scraped prices**

I. Introduction

Inflation measurement plays a central role in economic decision-making because it affects purchasing power, wage adjustments, household budgets, and public policy. In countries with high price volatility, such as Türkiye, changes in food, rent, transportation, and essential consumer goods are directly reflected in everyday life. Therefore, reliable and transparent inflation measurement is important not only for economic analysis but also for public understanding.

Traditional inflation measurement is commonly based on the Consumer Price Index (CPI; Turkish acronym: TUFE), which tracks price changes in a representative basket of goods and services [1]. Since households allocate different shares of their income to different consumption categories, CPI calculations use expenditure weights to reflect the relative importance of these categories. Recent computational approaches preserve this basket-based logic while applying it to price data collected from digital sources through web crawling and online price monitoring [2]–[5].

Although official CPI statistics provide the standard reference for inflation measurement, their data collection and aggregation procedures are not easily reproducible by independent researchers. This study proposes a transparent web-based framework that complements existing indicators by constructing a large-scale online price dataset for Türkiye, developing a crawling, cleaning, and price-normalization pipeline, and producing a CPI-weighted web-based price index together with a separate food-cost estimation framework. All datasets and codes used in this study are made publicly available.[1]

II. Literature Review

Inflation is commonly measured through a Consumer Price Index (CPI), which tracks price changes in a representative basket weighted by household consumption patterns. The international CPI framework defines Laspeyres-type indices as a practical standard and uses COICOP for expenditure classification [1]. In Türkiye, the official CPI is published monthly by TUIK using a COICOP-based basket and official price collection procedures [6]. Alongside official statistics, ENAG publishes an alternative high-frequency E-TUFE indicator [7], ITO reports Istanbul-level price statistics [8], and TURK-IS publishes monthly hunger and poverty threshold estimates for a four-person household [9].

Online prices have increasingly been used as complementary data sources for inflation measurement. Eurostat and the UK Office for National Statistics have examined web scraping for official price statistics [2], [3], while the Billion Prices Project demonstrated that online price indices can provide high-frequency information on price dynamics [4]. For Türkiye, Soybilgen, Yazgan, and Kaya constructed food-price subindexes from daily online prices, representing the closest Türkiye-focused predecessor to this study [5]. This study extends that line of work by collecting online prices across multiple consumption categories rather than focusing only on food.

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[1] https://github.com/urazkagangunes/InflationResearchStudy

III. Datasets

To construct a CPI-weighted web-based price index for the categories covered in this study, price data were collected from multiple online sources across several consumer domains. These domains include supermarkets, clothing stores, construction-supply markets, cosmetics retailers, home-goods sellers, technology retailers, and rental-housing listings. For each source, product names, prices, and available product attributes were extracted and standardized for further analysis.

A. Data Collection Process

Data were collected through a hybrid pipeline combining static HTML parsing, browser automation for JavaScript-heavy websites, and API access when available. The crawlers respected `robots.txt` instructions when available, used conservative request delays, and relied on `sitemap.xml` files where possible. For rental housing, a widely used real estate platform was selected because it provided broad geographic coverage and standardized listing information, although listings may reflect asking rather than realized transaction prices.

All records were stored in CSV format and converted into a common structure. Products were matched across dates using stable identifiers whenever available, including product URLs, product IDs, normalized names, and brand information. Row order was not used for matching because online listings may change dynamically. The main challenges were dynamic page structures, layout changes, temporary access failures, and the computational cost of daily collection.

B. Dataset Statistics and Cleaning

Excluding the auxiliary health-related category, the raw collected dataset comprises 34,203,627 price records and 568,204 unique items. Data collection started on 1 March 2026 and is ongoing; the analysis uses the subset collected up to 1 June 2026. Health-related products were collected but treated as auxiliary because their coverage was more limited and less continuous.

Table I reports unique items and raw data rows for the main product domains. The largest item counts were obtained from clothing stores, supermarkets, and construction-supply markets, reflecting differences in product diversity and online listing availability across categories.

Before inflation calculation, duplicate records, missing or non-numeric prices, malformed price fields, scraping artifacts, and extreme outliers were removed. Products unavailable at either endpoint of a comparison window were not imputed and were excluded from that window. Thus, each monthly estimate was based only on product–store pairs observed at both endpoint dates. Not all raw rows were used in the final inflation calculation, as the cleaned and matched dataset is smaller than the total raw row count reported in Table I.

TABLE I
Unique Items and Total Raw Data Rows by Main Product Category

Category	Unique	March	April	May	Total Rows
Rental Housing	13,061	3,024,424	2,325,182	1,700,071	7,049,677
Supermarkets	128,527	3,122,338	2,591,882	1,676,693	7,390,913
Clothing	140,297	2,649,010	3,024,944	2,098,934	7,772,888
Construction Supplies	104,936	2,322,492	2,983,961	2,130,045	7,436,498
Cosmetics	88,908	-	450,686	1,694,703	2,145,389
Home Goods	66,113	-	118,180	1,748,703	1,866,883
Technology	26,362	-	-	541,379	541,379
Total	**568,204**	**11,118,264**	**11,494,835**	**11,590,528**	**34,203,627**

Health-related products are excluded from this table owing to their auxiliary status; see Table V for category-level coverage details.

IV. Methodology

A. Inflation Calculation

The web-based CPI-weighted inflation estimate was calculated from endpoint-matched online price observations. For each comparison window, duplicate rows for the same product within a store were first collapsed by averaging their prices. Current and previous snapshots were then inner-joined using store, normalized product name, COICOP category, and sector. Therefore, only product–store pairs observed at both endpoints of the comparison window entered the calculation.

The price change for product j in store s was calculated as

$$r_{j,s,t} = 100 \left(\frac{p_{j,s,t}}{p_{j,s,t-1}} - 1 \right), \tag{1}$$

where $p_{j,s,t}$ and $p_{j,s,t-1}$ denote the current and previous prices, respectively. Product–store pairs with absolute price changes exceeding 80% were treated as potential scraping or unit-parsing artifacts and excluded from the calculation.

When the same product was observed in multiple stores, its store-level price changes were averaged using equal store weights. Products appearing in multiple sectors mapped to the same COICOP category were subsequently collapsed by simple averaging to prevent duplicate representation. The category-level change was calculated as the equal-weight arithmetic mean of the valid deduplicated product changes within that category.

The aggregate product-based index was calculated as

$$\pi_t = \sum_{c \in C_t} \widetilde{w}_{c,t} \bar{r}_{c,t}, \qquad \widetilde{w}_{c,t} = \frac{w_c}{\sum_{k \in C_t} w_k}, \tag{2}$$

where $\bar{r}_{c,t}$ denotes the average valid product-level price change in category c, w_c denotes its original CPI weight, and C_t denotes the categories with valid paired product observations in comparison window t. Categories without valid paired observations were excluded, and the weights of the remaining categories were re-normalized separately for each window.

Each monthly estimate was computed independently from its two endpoint snapshots rather than chained across months. Consequently, each window uses a window-specific matched panel, and the set of matched products, stores, and represented categories may differ across comparison windows. The resulting measure should therefore be interpreted as a CPI-weighted

web-based index for the covered product categories rather than a full reproduction of the official CPI.

Rental housing was calculated separately from the product categories. To maintain a consistent aggregation basis across all comparison windows, the principal aggregate reported in this paper is the product-based CPI-weighted index and excludes rental housing. City-level average rents were compared only for cities available on both dates, and the resulting rent change was reported separately as a category-level reference. For each common city, rent inflation was computed as the percentage change in mean listing price, and the overall rental indicator was obtained as the arithmetic mean of these city-level changes (Carli index), assigning equal weight to each city regardless of population or market size. Since rental listings may reflect asking prices rather than realized transaction prices, this component should be interpreted as an online rental-price indicator.

B. Hunger Threshold Calculation

The hunger threshold was estimated for a four-person household using a fixed basket of 16 food items across eight food categories. The basket items and monthly quantities were defined with dietitian support and aligned with the nutritional recommendations reported in the Turkish Dietary Guidelines (TUBER) [10]. Quantities were held constant across months so that changes in the estimated basket cost reflect price movements rather than changes in assumed consumption.

Prices were collected from 13 online supermarkets. Scraped products were assigned to basket items using transparent keyword and exclusion rules, and package sizes were extracted from product names to standardize prices as TRY per kilogram, liter, or piece. Within each supermarket, valid standardized prices matched to each basket item were summarized using mean, median, and minimum values. When multiple snapshots were available for the same supermarket and month, their item-level prices were first averaged to obtain one monthly observation per supermarket.

For each basket item, supermarket-level prices were then aggregated across the available supermarkets using equal supermarket weights. The monthly basket cost under aggregation strategy a was calculated as

$$H_t^{(a)} = \sum_{i=1}^{16} q_i p_{i,t}^{(a)}, \tag{3}$$

where q_i denotes the fixed monthly quantity of basket item i, $p_{i,t}^{(a)}$ denotes its representative cross-supermarket unit price in month t, and $a \in \{\text{mean}, \text{median}, \text{minimum}\}$.

A monthly estimate was reported only when all 16 required basket items had valid prices; incomplete months were flagged rather than silently summed or imputed.

The basket is defined at the aggregate household level and does not model separate age- and sex-specific sub-baskets. Therefore, differences between the proposed estimates and TURK-IS may reflect both basket composition and price-aggregation choices. Following the TURK-IS framework, food expenditure may also be used as an input for broader poverty-threshold estimates by assuming that it corresponds to 30.70% of total household expenditure [9]. The resulting values should therefore be interpreted as complementary cost-of-living indicators rather than exact replications of the TURK-IS calculation.

TABLE II
EXAMPLE FOR MAPPING BASKET ITEMS TO SCRAPED PRODUCT NAMES

Basket Item	**Target Unit**	**Matched Scraped Examples**	**Aggregation Input**
Chickpeas	Kg	Nohut 1000 GR; Nohut 2000 GR	TRY per kg
Bread	Kg	Ekmek 500 G; Ekmek 800 GR; Roll ekmek 50 GR	TRY per kg
Olive Oil	Liter	Sizma zeytinyagi 1000 ml	TRY per liter
Grissini	Kg	Kepekli grissini 175 GR; Sade grissini 175 GR	TRY per kg

C. Product Matching and Price Normalization

Products were matched using a rule-based procedure based on normalized product names, product-type keywords, package-size expressions, and measurement units when available. Product names were lowercased, special characters were removed, whitespace was standardized, and measurement units such as "1 LT," "1L," and "1000 ML" were transformed into a common format. Products that could not be matched reliably were excluded to reduce artificial price differences.

For food and supermarket products used in the hunger-threshold analysis, prices were converted to standardized unit prices whenever package-size information was available. Non-food categories such as clothing, technology, rental housing, cosmetics, construction supplies, and home goods were analyzed using cleaned nominal product prices after product matching. Where multiple sectors map to the same TUIK category — specifically, home goods and construction supplies are both assigned to group 05 — products matched across those sectors are further deduplicated by averaging, so that no physical product is counted more than once in the category-level index.

Table II summarizes how nutrition-basket items were mapped to observed online product names before price aggregation. The basket item defines the target product group and required unit, while the scraped product names provide candidate observations. Candidate products were retained when their normalized names and package-size expressions were compatible with the basket item; otherwise, they were excluded from the basket-level calculation.

Each basket item may correspond to multiple observed product names and package sizes. These records are mapped to the same basket item, normalized to a common unit, and then aggregated using mean, median, and minimum standardized prices.

Fig. 1 illustrates the unit-normalization logic using an anonymized milk example from three markets. Observed package prices are converted to a common 1-liter equivalent before representative prices are computed.

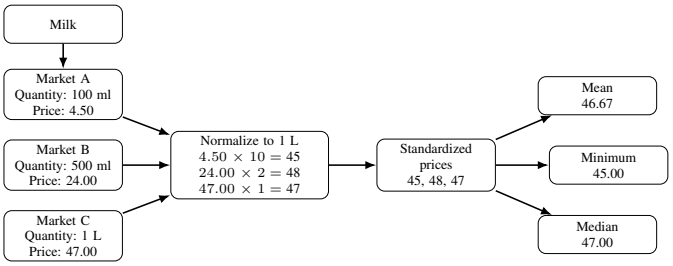


Fig. 1. Schema for Milk Price Normalization Across Three Anonymous Markets

TABLE III
MONTHLY INFLATION ESTIMATES COMPARED WITH REFERENCE SOURCES (%). TUIK FIGURES: [11]–[13]; ENAG FIGURES: [14]–[16]; ITO FIGURES: [17]–[19].

Source	March	April	May
TUIK (official CPI)	1.94	4.18	1.71
ENAG (independent)	4.10	5.07	2.16
ITO (Istanbul)	2.97	3.74	1.53
This study	**2.00**†	**1.16**	**0.62**‡

Study values correspond to the product-based TUIK-weighted index; rental housing is excluded from the aggregate and reported separately.

† Rental housing excluded: only 1 common city (Kırşehir) available; figure reflects TUIK-weighted estimate without rent component.

‡ Rental housing excluded: only 3 geographically clustered cities available; figure reflects TUIK-weighted estimate without rent component. June data also incomplete.

The conversion used in Fig. 1 follows

$$p_i^{std} = p_i \times \frac{1}{q_i}, \tag{4}$$

where p_i is the observed package price and q_i is the package quantity expressed in liters. For example, a 100 ml package has $q_i = 0.10$, so a price of 4.50 corresponds to 45.00 per liter. After standardization, mean, median, and minimum prices are reported as alternative representative prices.

V. EXPERIMENTS

A. Comparison with TUIK, ENAG, and ITO

Table III compares the proposed estimates with TUIK, ENAG, and ITO. The windows ending on 1 April, 1 May, and 1 June represent March, April, and May, respectively, although they do not exactly match official calendar-month boundaries.

The product-based estimates were 2.00%, 1.16%, and 0.62% for March, April, and May, respectively. The March estimate slightly exceeded TUIK's 1.94% but remained below ENAG's 4.10% and ITO's 2.97% [11], [14], [17]; rental housing was excluded because only Kırşehir was common to both endpoint dates. The April estimate was below TUIK, ENAG, and ITO [12], [15], [18]; the separate rental indicator increased by 1.62% across 21 common cities but was not included in the aggregate. Utility tariff changes affecting official April inflation were also not captured by the online product data. The May estimate remained below all three references [13], [16], [19]; rental housing was excluded because only three geographically clustered cities were available and June coverage was incomplete.

TABLE IV
TUIK COICOP BASKET WEIGHTS AND FULL-COVERAGE NORMALIZED WEIGHTS.

Category	CO.	TUIK (%)	Normalized (%)
Food & non-alcoholic bev.	01	24.44	39.39
Clothing & footwear	03	7.90	12.73
Housing & utilities	04	11.40	18.37
Furnishings & home equip.	05	7.92	12.76
Health	06	2.79	4.50
Information & comm.	08	3.10	4.99
Personal care & misc.	13	4.49	7.24
Covered subtotal		**62.04**	**100.00**
Rest. & accommod. (no data)	*11*	*11.13*	—
Other not covered		*26.83*	—

TABLE V
NUMBER OF STORES AND UNIQUE PRODUCTS PER COICOP CATEGORY BY COLLECTION DATE

		1 Mar		1 Apr		1 May		1 Jun	
Category	**CO.**	*S*	*P*	*S*	*P*	*S*	*P*	*S*	*P*
Food	01	13	61,585	11	59,149	10	52,975	3	9,608
Clothing	03	–	–	9	41,349	9	43,109	3	7,472
Home & Const.	05	–	–	11	56,140	14	81,088	4	27,685
Health	06	1	13,707	1	13,694	2	13,739	–	–
Rental Housing	04	9	–[a]	27	–[a]	21	–[a]	3	–[a]
Technology	08	–	–	–	–	2	515	2	1,423
Cosmetics	13	–	–	–	–	7	42,533	2	12,242
Total	–	**14**	**75,292**	**32**	**170,332**	**44**	**233,959**	**14**	**58,430**

[a] For rental housing, *S* denotes city-level rental coverage and is excluded from the total store count; product count is not applicable. *S*: stores; *P*: deduplicated products. Dashes indicate that no comparable data were available for that category on the corresponding date.

B. Category-Based Inflation Results

Table IV presents the official TUIK COICOP basket weights and their normalization over the full set of categories potentially covered by the dataset. The covered categories collectively account for 62.04% of the full TUIK basket weight as of 1 May 2026. For the reported product-based aggregate, weights were re-normalized separately in each comparison window over categories with valid product observations; rental housing was excluded and reported separately. COICOP group 11 (Restaurants and accommodation services, weight 11.13%) did not have sufficient paired observations on the monthly reference dates and therefore contributed no observations to the reported estimates.

Table V presents the number of stores and unique products per TUIK COICOP category across all collection dates, providing an overview of dataset growth and coverage over the observation period.

The dataset grew substantially from March to May, with total deduplicated products increasing from 75,292 to 233,959 and store count rising from 14 to 44. The June collection shows a marked contraction across all categories, with total stores dropping back to 14 and products to 58,430. This reduction reflects the early stage of June data collection rather than a structural change in coverage, and is expected to recover as collection continues.

TABLE VI
CATEGORY-LEVEL AVERAGE INFLATION BY MONTHLY WINDOW (%).

Category	CO.	Mar	Apr	May
Food (supermarkets)	01	2.22	1.84	1.26
Clothing	03	–[a]	−1.34	−0.42
Home goods & construction	05	–[a]	1.95	0.42
Health	06	≈0.00[b]	≈0.00	–
Rental housing	04	−1.59[c]	1.62	2.06[c]
Technology	08	—[d]	—[d]	—[d]
Cosmetics	13	—[d]	—[d]	−0.69
Aggregate (TUIK-weighted)	—	**2.00**[c]	**1.16**[c]	**0.62**[c]

[a] No reliable monthly paired data available for March window; category excluded from March estimate.
[b] Health-related prices remained unchanged between snapshots (0.00% on 13,662 matched pairs in March); included in the aggregate at its normalized weight.
[c] The reported aggregate excludes rental housing in all windows to maintain a consistent product-based index. Rental figures are shown separately; the March and May estimates additionally have limited common-city coverage (1 and 3 cities, respectively).
[d] Category entered the dataset late or had insufficient paired observations for a reliable standalone category estimate.

Table VI presents the web-based inflation estimates disaggregated by product category for the three monthly date-to-date windows. Category-level figures are reported as rounded arithmetic means of valid deduplicated product price changes within each TUIK COICOP group. The aggregate row reports the product-based CPI-weighted index calculated from the corresponding full-precision category values. For each window, categories without valid paired product observations were excluded and the remaining CPI weights were re-normalized. Rental housing is reported separately and is not included in the aggregate.

Among the tracked categories, rental housing showed a reported change of −1.59% in the first window; however, this figure is based on a single city (Kırşehir) and is excluded from the March aggregate. For the April window, rental housing recorded a reliable 1.62% increase across 21 cities. The May rent figure of 2.06% is based on 3 cities in eastern Anatolia and is similarly excluded from the May aggregate.

Supermarket food prices showed consistent upward pressure across all three periods (2.22%, 1.84%, and 1.26%), in line with the broader food inflation trend observed by official sources. Home goods and construction supplies recorded a moderate increase of 1.95% in the April window, declining to 0.42% in May. Clothing prices declined across the April and May windows (−1.34% and −0.42%), consistent with seasonal discount campaigns typical of the spring and early summer retail period.

Health-related prices remained essentially unchanged across the matched observations. In the March window, Health was matched on 13,662 product-store pairs with a measured change of 0.00%, and is included in the aggregate at its normalized weight; this partly explains why the March aggregate (2.00%) falls below the unweighted Food-only rate (2.22%). Nevertheless, the health category should be interpreted cautiously because its coverage is narrower and less continuous than the main product domains. Technology and cosmetics categories entered the dataset from May 2026. Cosmetics recorded a decline of −0.69% in May; technology data were insufficient for a monthly estimate in the available windows.

TABLE VII
MONTHLY FOOD BASKET USED FOR HUNGER-THRESHOLD ESTIMATION

Category	Product	Unit	Quantity
Dairy Products	Yogurt	Kg	59.7
Dairy Products	White Cheese	Kg	5.7
Meat and Protein	Cubed Meat / Lamb Meat	Kg	4.6
Meat and Protein	Chicken	Kg	10.3
Meat and Protein	Fish	Kg	6.9
Meat and Protein	Eggs	Piece	120.0
Legumes	Chickpeas	Kg	1.8
Nuts and Seeds	Walnut / Hazelnut / Peanut	Kg	2.7
Grains	Bread	Kg	18.0
Fruits	Banana	Kg	16.7
Fruits	Seasonal Fruit	Kg	12.9
Vegetables	Onion	Kg	18.0
Vegetables	Eggplant / Zucchini	Kg	23.1
Vegetables	Other Vegetables	Kg	11.8
Oils	Olive Oil	Liter	1.1
Other Food Products	Grissini	Kg	2.1

TABLE VIII
COMPARISON OF HUNGER THRESHOLD ESTIMATES

Month	Mean Est.	Median Est.	Min. Est.	TURK-IS
February 2026	48,962	39,970	25,809	32,365
March 2026	47,099	38,942	23,915	32,792
April 2026	46,644	38,797	22,777	34,586
May 2026	51,562	42,844	25,868	35,174

C. Hunger Threshold Results

The hunger-threshold estimates were compared with the monthly values published by TURK-IS for a four-person household. Since the proposed calculation relies on web-scraped online prices and a nutrition-plan-based food basket, it should be interpreted as a complementary cost-of-living reference rather than an exact replication of the TURK-IS methodology. Although the main multi-category inflation dataset begins in March 2026, the hunger-threshold comparison additionally includes February 2026 observations available for the nutrition-basket data.

The food basket was prepared with dietitian support and aligned with nutritional recommendations reported in the Turkish Dietary Guidelines (TUBER) [10]. The basket used for the hunger-threshold estimation is reported in Table VII.

Table VIII compares the proposed estimates with TURK-IS values reported for February, March, April and May 2026 [9], [20]–[22]. All values are reported in Turkish lira (TRY).

Across the four examined months, mean-based estimates exceeded the corresponding TURK-IS values by 34.86%–51.28%, while median-based estimates were 12.18%–23.50% higher. Minimum-price estimates were 20.26%–34.15% lower and should therefore be interpreted as lower-bound benchmarks. These differences reflect the sensitivity of the mean to high-priced products, the greater robustness of the median to extreme observations, and the assumption underlying the minimum strategy that consumers can consistently select the least expensive qualifying products. Month-to-month variation may additionally reflect food-price dynamics, promotions, retailer-specific pricing, and seasonal availability.

VI. Discussion and Conclusion

This study presented a CPI-weighted web-based framework for monitoring online price dynamics and cost-of-living changes in Türkiye. The reported aggregate is a product-based CPI-weighted index that excludes rental housing in all three comparison windows, while rental changes are presented separately as category-level references. The March and May rental estimates have limited common-city coverage, whereas the April rental estimate is based on 21 common cities. The April product-based aggregate remains lower than official and alternative indicators partly because utility tariff changes are not directly captured by the online product data.

The hunger-threshold results indicate that nutrition-based web price monitoring can also serve as a complementary cost-of-living reference. Mean- and median-based estimates remain above TURK-IS values, while minimum-price estimates remain below them and should be interpreted as lower-bound benchmarks. None of these estimates should be interpreted as direct replacements for TURK-IS because basket definitions, retailer coverage, and pricing procedures differ.

The findings also show that web-based price monitoring is most useful as a high-frequency supplementary signal rather than as a standalone inflation measure. The framework can reveal category-specific movements, coverage gaps, and online-market dynamics that may not be visible from aggregate monthly indicators alone. Therefore, it is valuable for exploratory monitoring, early detection of price changes, and transparent comparison across data sources, while still requiring careful interpretation when online coverage is incomplete or category composition differs from official CPI baskets.

The main contribution of this study is a reproducible pipeline for collecting, cleaning, matching, weighting, and analyzing online prices across multiple consumer domains. Limitations include partial CPI-category coverage, changing source and product composition across comparison windows, possible online–offline price differences, regional variation, temporary promotions, stock changes, and website instability. Although each monthly estimate uses product–store pairs observed at both endpoints, a single fixed product panel is not maintained across all months. The current analysis does not report bootstrap uncertainty intervals, systematic sensitivity analyses for alternative outlier thresholds, or formal manual validation statistics for the rule-based matching procedure. Rental cities and supermarkets are aggregated with equal weights, without adjustment for population, market size, or market share. Future work will examine stable-source panels, alternative outlier settings, manually validated product matches, and median- or composition-adjusted rental indices.

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